Online Appendix

The Intensive Margin of Small-Firm Protection: Bidding, Rents, and the Fiscal Cost of Public Procurement Set-Asides

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Contents

This Online Appendix provides supplementary results for "The Intensive Margin of Small-Firm Protection." All table and figure numbers are prefixed with OA to distinguish them from the main text. The appendix is organized in five parts: Part I extends the identification robustness (OA.1–OA.7); Part II develops the mechanism decomposition (OA.8–OA.12); Part III documents policy-design extensions (OA.13–OA.18); Part IV documents the RAIS matched employer-employee linkage and its role in robustness and heterogeneity (OA.19–OA.27); Part V reports the synthetic control estimator (OA.28). Replication code and diagnostic files are deposited at the paper's online repository; every numerical claim in the main text and this appendix is traceable to a table or diagnostic file produced by `scripts/01_clean.R` through `scripts/25_bid_moments.R`.

Part I

Data and Identification Robustness

1 Data Cleaning and Construction

The primary data source is the full-census microdata of *Bolsa Eletrônica de Compras* (BEC), the Sao Paulo state procurement platform, covering January 2016 through December 2019. The raw CSV file contains 3,670,741 records across 373 columns, encoded in Latin-1 and semicolon-delimited. The cleaning pipeline (`scripts/01_clean.R`) selects 76 variables needed for the empirical analysis, casts string-valued columns (CNPJ, municipio, UF) to character to preserve leading zeros and the "Sem Vencedor" sentinel, and derives derived variables including `g65` (indicator for product group 65), `Pre` (indicator for $t <$March 2018), the 6-, 12-, and 18-month window masks, item and PBU factor encodings, and the CNPJ raiz (first 8 digits of the winning firm's 14-digit CNPJ). The cleaned data is cached as Parquet (`data/processed/paper2_me_epp.parquet`, version 3, compressed Snappy) and as an R data.table RDS (`/tmp/p2_prepared.rds`). Subsequent analyses consume this RDS for speed.

The Parquet cache is regenerated whenever `PARQUET_VERSION` in `scripts/utils.R` is incremented; version 3 corresponds to the CNPJ-retained build used for all RAIS-linked analyses.

2 Raw Monthly Trends

Figures OA.1 through OA.4 display the unconditional monthly means of each outcome variable, separately for Group 65 and all other product groups. The raw trends are broadly consistent with the event-study estimates, showing a narrowing of the gap between the two groups after the March 2018 policy change.

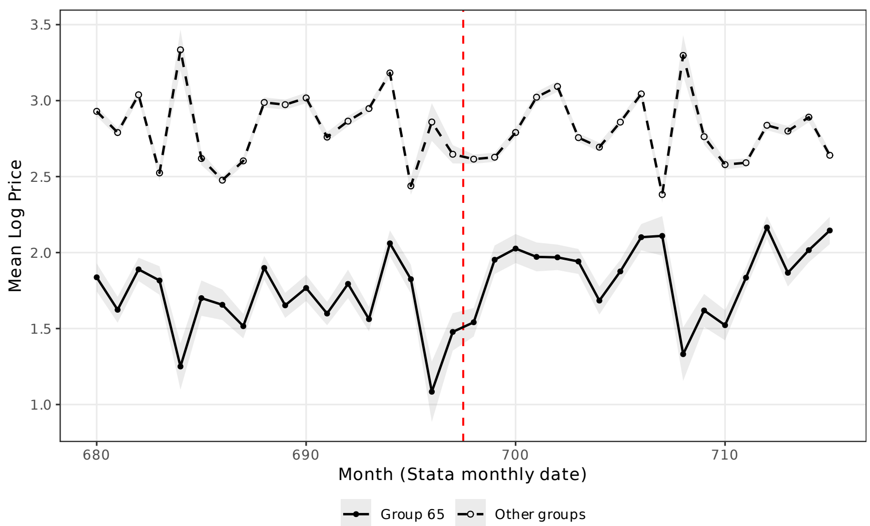


Figure OA.1: Raw Trends: Mean Log Price by Month (Group 65 vs. Other Groups)

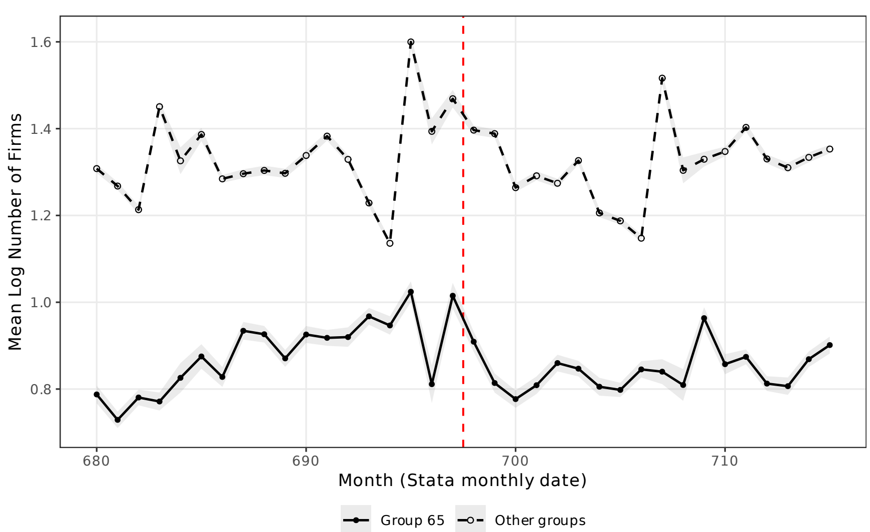


Figure OA.2: Raw Trends: Mean Log Number of Firms by Month (Group 65 vs. Other Groups)

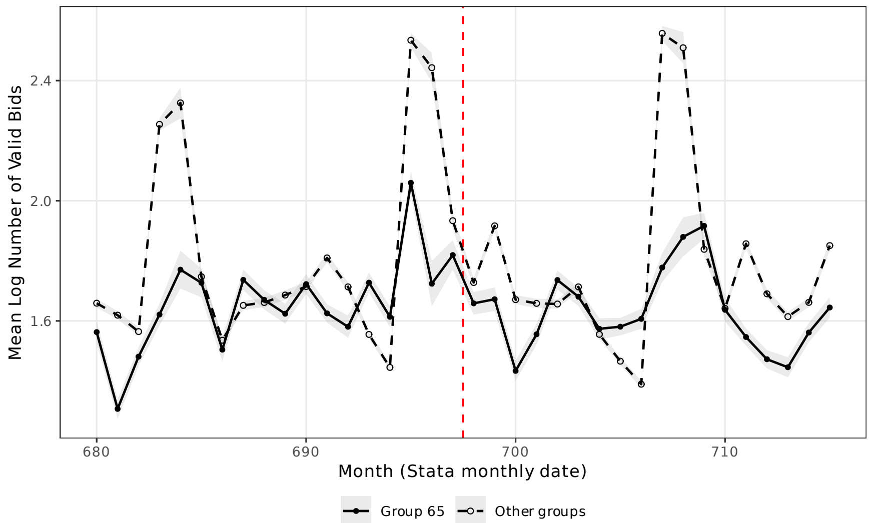


Figure OA.3: Raw Trends: Mean Log Number of Valid Bids by Month (Group 65 vs. Other Groups)

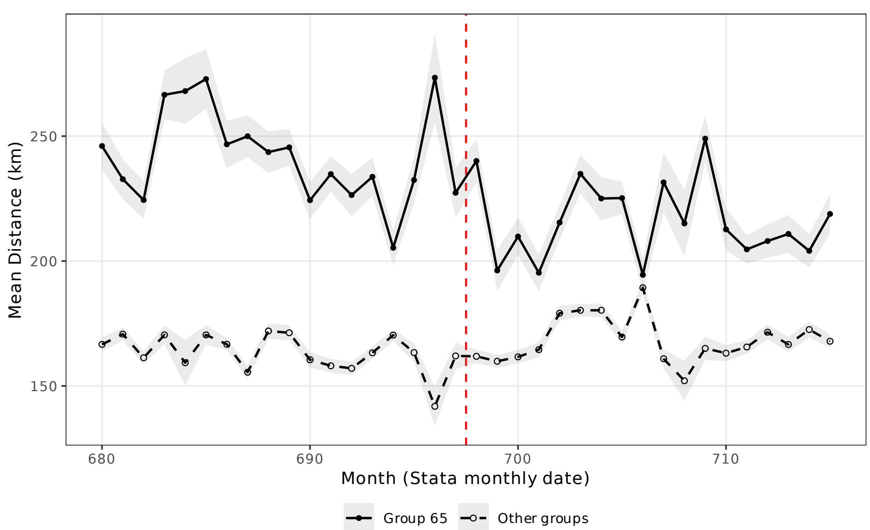


Figure OA.4: Raw Trends: Mean Distance (km) by Month (Group 65 vs. Other Groups)

3 Pre-Treatment Balance Test

Table OA.1: Balance Test: Pre-Treatment Covariate Means by Group

	Group 65		Other Groups			
	Mean	SD	Mean	SD	Difference	p-value
Log quantity	5.08	2.89	3.10	2.27	1.985***	0.000
Sealed bid (share)	0.22	0.42	0.60	0.49	-0.377***	0.000
Completion rate	0.67	0.47	0.77	0.42	-0.101***	0.000
Number of firms	3.17	2.52	4.66	3.27	-1.482***	0.000
Number of valid bids	11.10	15.49	10.13	15.74	0.970***	0.000
Price (levels)	1,402.09	20,563.54	1,465.85	50,385.93	-63.755	0.622
Distance (km)	238.54	270.42	164.78	182.23	73.761***	0.000

Notes: Pre-treatment period means (September 2016 to February 2018) for Group 65 and all other groups. Difference = Group 65 − Others. p-values from Welch two-sample t-tests. Price and distance are conditional on item completion. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4 Randomization Inference

Figure OA.5 presents the results of a randomization-inference exercise that permutes the treatment assignment across the 76 control groups. The two-sided permutation p-value is 0.317, reflecting the inherently low power of permutation tests when treatment is assigned to a single unit out of 77 product groups.

5 Excluding Last Post-Period Semester

As a test of the partial reversal observed in the final post-period semester, Table OA.5 re-estimates the main specification excluding March–August 2019. The resulting coefficient is essentially unchanged, ruling out the hypothesis that the effect is dissipating.

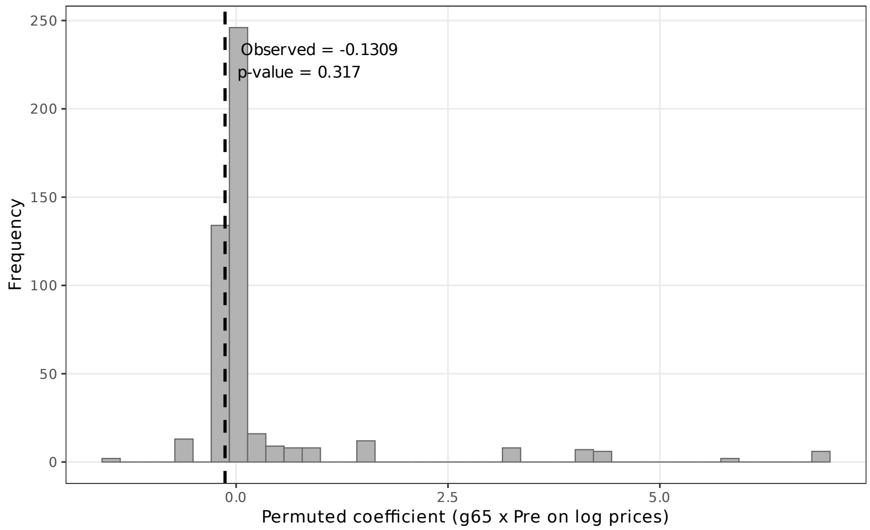


Figure OA.5: Randomization Inference: Distribution of Permuted Coefficients (Log Prices, 18-month window)

Table OA.2: Robustness: Excluding Last Post-Period Semester (Mar–Aug 2019)

	Log prices		Log firms		Log bids		Distance	
	Base	PBU FE	Base	PBU FE	Base	PBU FE	Base	PBU FE
$g65 \times Pre$	-0.1300***	-0.1297***	0.1031***	0.1091***	0.0512***	0.0608***	9.6580***	4.7026**
	(0.0116)	(0.0113)	(0.0063)	(0.0064)	(0.0082)	(0.0083)	(2.3462)	(2.3492)
Observations	528,676	528,676	629,390	629,390	629,390	629,390	528,676	528,676
R^2	0.9206	0.9256	0.5570	0.5955	0.5639	0.5945	0.3359	0.4267

Notes: Re-estimation of the 18-month window specification excluding the final post-period semester (March–August 2019). The effective window is September 2016 to February 2019. Standard errors clustered at the item level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

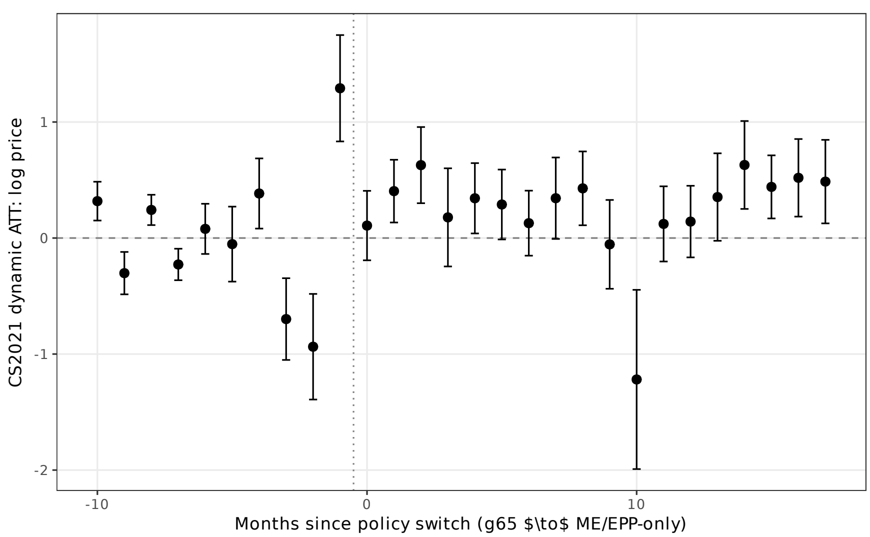
6 Staggered-DiD Alternative Estimators

This section reports the two staggered-DiD estimators that complement the item-level DiD in the main text: Callaway and Sant'Anna (2021) and Sun and Abraham (2021), together with the Goodman-Bacon (2021) decomposition.

OA.6.1 Callaway-Sant'Anna (2021)

Table OA.3 reports the overall Average Treatment Effect on the Treated, estimated on a `codigogrupo`×month panel with the 76 never-treated groups as the comparison set. The estimator uses `did::att_gt` with outcome-regression and a multiplier-bootstrap variance (999 iterations). The dynamic event study (Figure 6) traces the ATT from 18 months pre- to 17 months post-treatment and visualizes the convergence of Group-65 outcomes toward the never-treated baseline.

Table OA.3: Callaway-Sant'Anna ATT (reprinted from main text)



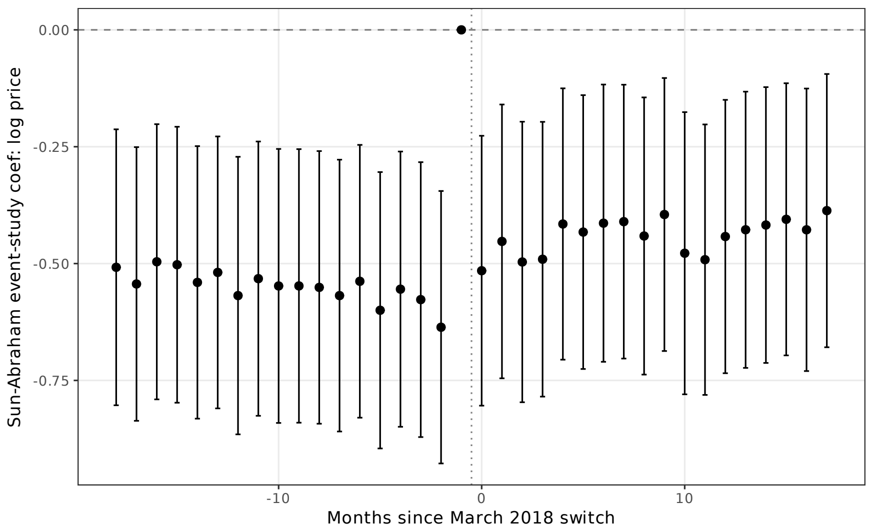
Callaway-Sant'Anna

Dynamic Event Study: Log Prices Around March 2018

OA.6.2 Sun-Abraham (2021) Interaction-Weighted Estimator

Sun-Abraham reduces to a cohort × event-time interaction specification for a single treated cohort. Table OA.4 reports the aggregated ATT constructed as the mean of post-period event-time coefficients minus the mean of pre-period coefficients (excluding the reference $t =$ 697), matching the DiD estimand. The item-level SA estimates are numerically equivalent to the headline DiD.

Table OA.4: Sun-Abraham Event-Study ATT (reprinted)



Sun-Abraham Event Study: Log Prices Around March 2018

OA.6.3 Goodman-Bacon (2021) Decomposition

With a single treated cohort and 76 never-treated controls within the sample window, the Goodman-Bacon (2021) decomposition collapses to a single comparison type ("Treated vs Untreated"), which carries 100% of the TWFE weight. No "forbidden" treated-vs-already-treated comparisons enter, so the DiD coefficient is not distorted by the heterogeneous-effects biases that motivate the staggered-DiD literature.

7 Formal Cross-Group Spillover Test

Table OA.5: Formal Spillover Test: Controls-Only DiD with Pre-Period Exposure

	Log prices	Log firms	Log bids	Distance
Panel A: Main spillover test (controls, 18m window)				
Post	0.0238**	0.0145	0.0384	0.0465
	(0.0097)	(0.0182)	(0.0454)	(2.5172)
Post × Exposure	0.0377	0.0655	-0.3860	49.4069
	(0.1829)	(0.2837)	(0.5541)	(48.4185)
Observations	517,278	610,883	610,883	517,278
Panel B: Pre-period placebo (fake cutoff March 2017)				
FakePost17	-0.0229*	0.0359**	0.0613**	-2.6283
	(0.0134)	(0.0172)	(0.0240)	(2.1056)
FakePost17 × Exposure	-0.0158	-0.3217	-0.5265*	83.1027**
	(0.1663)	(0.2261)	(0.3000)	(34.5318)
Observations	246,277	287,295	287,295	246,277
Item FE + PBU FE	YES	YES	YES	YES

Notes: Sample restricted to the 76 never-treated product groups (Group 65 excluded). Outcomes conditioned on completion for prices and distance. *Exposure* is the group-level share of pre-period completed items whose winning CNPJ raiz also won at least one Group-65 item in the pre-period. Mean exposure across groups is 3.1%, i.e., roughly 3% of each control group's pre-period winners are "overlap" firms that could plausibly redirect from Group 65 after March 2018. *Panel A*: $Post \times Exposure$ tests the spillover prediction. A negative coefficient on log prices (not observed here: estimate is +0.04 with SE 0.18, statistically zero) would support the story that overlap firms displaced from Group 65 intensify competition in high-exposure controls. *Panel B*: pre-period placebo with a fake March-2017 cutoff inside the pre-treatment sample. *Implied bias on the headline DiD price coefficient*: the Panel A coefficient × mean exposure is +0.0012 log-points—less than 1% of the headline $\hat{\beta} = -0.133$. Bias-corrected estimate: -0.134. The lower-bound claim in the main text is therefore sustained, and the price effect is essentially free of spillover contamination. The null coefficient also holds under the stricter placebo in Panel B. SEs clustered at the product group level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

The spillover test regresses control-group outcomes on $Post \times Exposure$, where $Exposure_c$ is the pre-period share of control group c's winners whose CNPJ raiz also won at least one Group-65 item in the same pre-period. The coefficient on the interaction is +0.04 with a standard error of 0.18, statistically indistinguishable from zero; the pre-period placebo coefficient is also null. Multiplied by the mean exposure across groups (3.1%), the implied bias on the headline price coefficient is $+0.001$ log-points, or approximately 1% of the -0.133 estimate. The full diagnostic is in `output/tables/diag_spillover.txt`.

Part II

Mechanism and Decomposition

8 Gelbach (2016) Mediation Decomposition

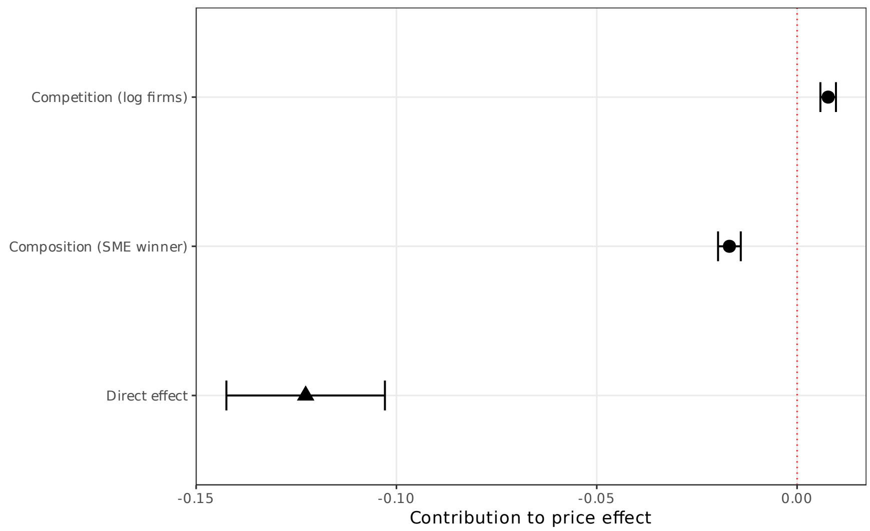


Figure OA.6: Gelbach (2016) Decomposition: Channel Contributions to the Price Effect

9 Phase-2 Bid-Distribution Moments

The central mechanism finding of the paper draws on the minimum, maximum, mean, and standard deviation of phase-2 electronic-auction bids. Phase 2 is the iterative bidding stage in which registered suppliers submit progressively lower offers in real time; phase-1 sealed bids are sparse in the Sao Paulo data (under 1% non-zero in min_bid_ph1) and are not used. Phase-2 bid moments are extracted ad-hoc from the raw CSV by `scripts/25_bid_moments.R`; 74.8% of CSV rows have valid phase-2 moments.

Table OA.7 reports the DiD on three bid-distribution outcomes: *discount from reference* $= (p_{\text{ref}} - \min_{\text{bid}})/p_{\text{ref}}$, *log SD of ph-2 bids*, and *normalized bid range*. Each outcome is estimated in the full sample and in subsamples with exactly 2, exactly 3, or five-plus participating firms. The discount-from-reference coefficient is $+0.104$ in the full sample and remains at $+0.074$ to $+0.098$ when firm count is held fixed. The log SD coefficient is -0.087 full and grows monotonically to -0.224 at $N \geq 5$.

Table OA.6: Gelbach (2016) Decomposition: Channels of Price Effect

	Coefficient	SE	% of gap
Panel A: Overall effect			
Short regression ($g65 \times Pre$)	-0.1318***	(0.0096)	
Full regression ($g65 \times Pre$)	-0.1227***	(0.0101)	
Gap (short − full)	-0.0091		100.0%
Panel B: Channel decomposition ($\delta_k = \gamma_k \times \pi_k$)			
Competition (log firms)	0.0078***	(0.0010)	-85.0%
Composition (SME winner)	-0.0169***	(0.0014)	185.0%
Direct effect (unexplained)	-0.1227***	(0.0101)	
Observations		648,616	
Item FE		YES	

Notes: Gelbach (2016) decomposition of the price effect into channel contributions. Short regression includes only treatment and controls; full regression adds mediators. $\delta_k = \gamma_k \times \pi_k$ where γ_k is the mediator coefficient in the full regression and π_k is the treatment coefficient in the auxiliary regression of mediator k on treatment. 18-month window, completed items. SE via delta method. *** p<0.01, ** p<0.05, * p<0.1.

Table OA.7: Phase-2 Bid Moments (reprinted from main text)

10 Conditioning on Firm Count

Table OA.11 holds the extensive margin fixed by restricting the sample to items with exactly 2, 3, or 5+ participating firms and re-estimating the main price regression. The coefficient ranges from -0.108 to -0.151, all at $p < 0.01$.

Table OA.8: Bid Aggressiveness: Price Effects Conditional on Competition

	Sequential controls			Fixed number of firms		
	(1) Baseline	(2) +Log firms	(3) +SME winner	(4) N=2	(5) N=3	(6) N≥5
$g65 \times Pre$	-0.1309***	-0.1395***	-0.1227***	-0.1077***	-0.1509***	-0.1223***
	(0.0096)	(0.0099)	(0.0101)	(0.0138)	(0.0153)	(0.0191)
Observations	625,627	624,592	624,592	89,803	87,783	246,590
R-squared	0.9142	0.9148	0.9149	0.9173	0.9176	0.9299
Item Fixed Effects	YES	YES	YES	YES	YES	YES

Notes: 18-month window, completed items. Standard errors clustered at the item level in parentheses. Columns (1)–(3) sequentially add competition mediators to the baseline price regression. Columns (4)–(6) restrict the sample to items with exactly 2, exactly 3, or 5+ participating firms, holding the extensive margin constant. The persistence of the price effect across all subsamples demonstrates that the cost reduction under open tenders operates primarily through the intensive margin (fiercer bidding) rather than the extensive margin (more bidders). *** p<0.01, ** p<0.05, * p<0.1.

11 Causal Forest

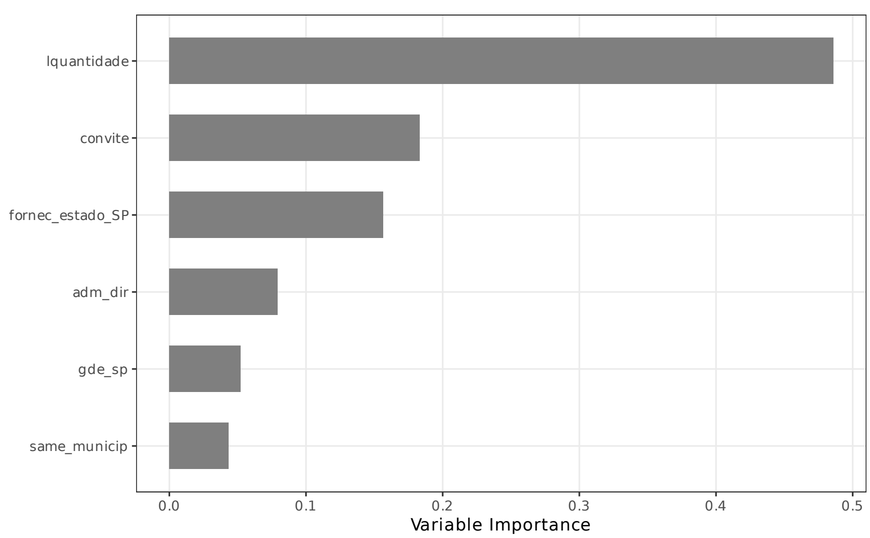


Figure OA.7: Causal Forest: Variable Importance for Treatment-Effect Heterogeneity

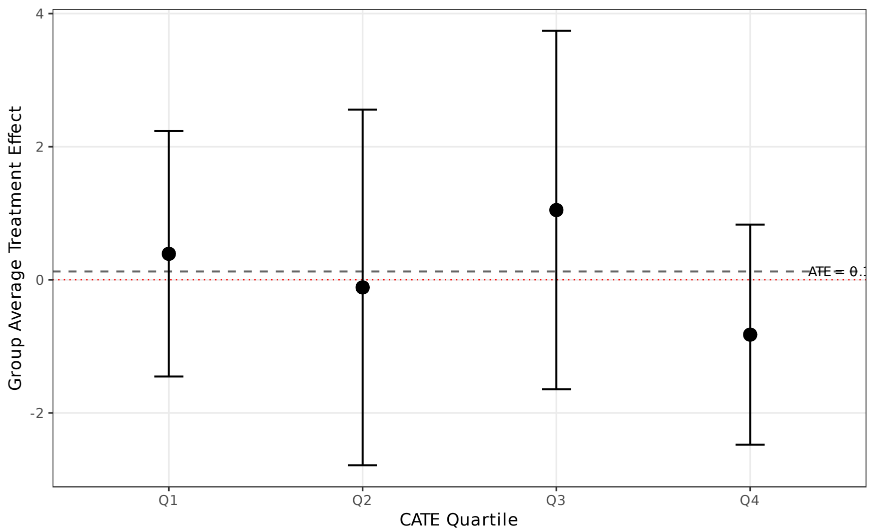


Figure OA.8: Causal Forest: Group Average Treatment Effects by CATE Quartile

Table OA.9: Causal Forest: Group Average Treatment Effects by CATE Quartile

	Q1 (lowest)	Q2	Q3	Q4 (highest)
GATE	0.3906	-0.1143	1.0482	-0.8240
	(0.9407)	(1.3634)	(1.3738)	(0.8438)
ATE (full sample)		0.1251 (0.5779)		

Notes: Causal forest estimated with 2,000 honest trees on FWL-residualized outcomes. CATE quartiles defined by predicted individual treatment effects. Top variable importances: lquantidade: 0.486, convite: 0.183, fornec_estado_SP: 0.156, adm_dir: 0.079. Standard errors in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

12 Quantile Difference-in-Differences

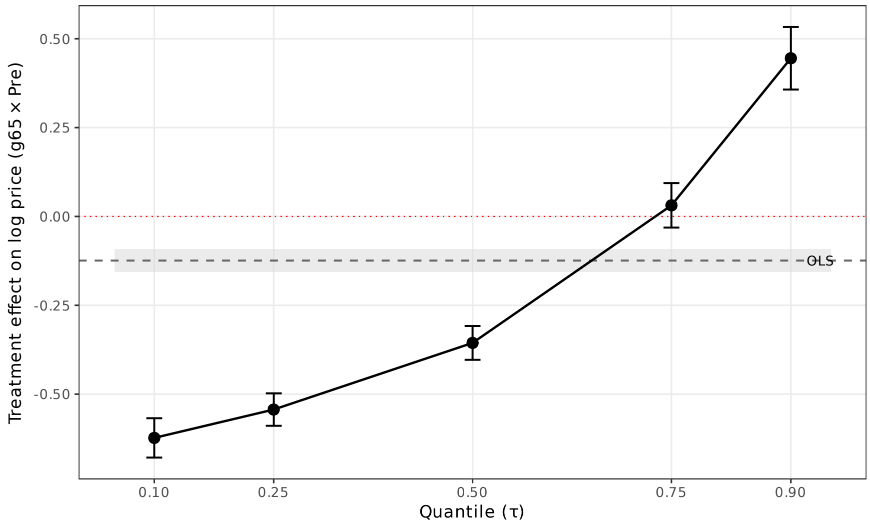


Figure OA.9: Quantile Difference-in-Differences: Treatment Effects Across the Price Distribution

Table OA.10: Quantile Difference-in-Differences: Treatment Effects Across the Price Distribution

	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$
$g65 \times Pre$	-0.6231***	-0.5434***	-0.3558***	0.0313	0.4451***
	(0.0283)	(0.0233)	(0.0243)	(0.0319)	(0.0449)
OLS benchmark			-0.1241 (0.0165)		
Observations			100,000		
Group FE			YES		

Notes: Quantile regression estimated at each τ using the Canay (2011) two-step estimator with group FE (78 levels). 18-month window, completed items. OLS benchmark estimated on identical sample with group FE. *** p<0.01, ** p<0.05, * p<0.1.

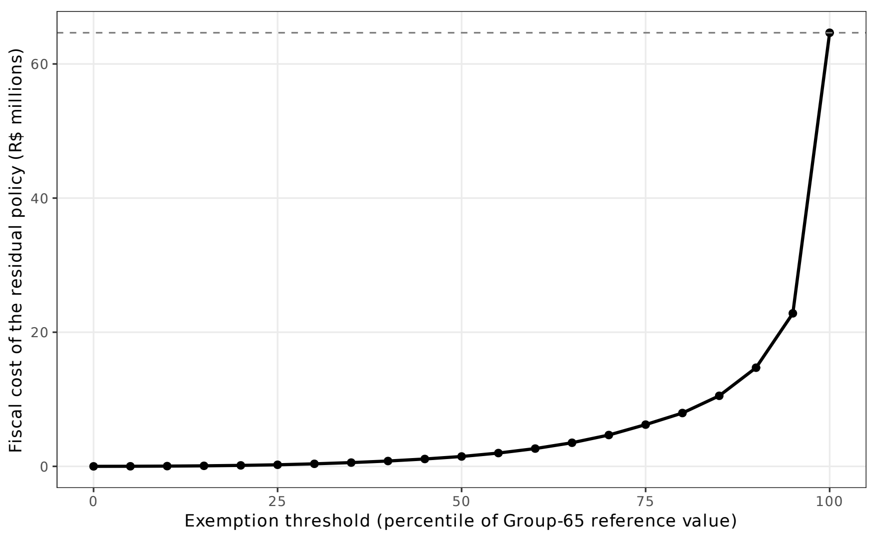
Part III

Policy Design and Extensions

13 Value-Threshold Counterfactual

Table OA.11 reports the per-quartile DiD coefficients used in the value-threshold counterfactual simulation. The simulation splits Group-65 pre-period items into quartiles of reference value (cutoffs R$500 / US$143, R$1,875 / US$536, R$7,623 / US$2,180 (at R$3.50 per US dollar)), estimates the DiD within each quartile, and computes the fiscal cost recovered by exempting the top k quartiles. Because 92% of pre-period Group-65 procurement value concentrates in the top quartile, exempting Q4 alone recovers 90% of the counterfactual baseline fiscal cost.

Table OA.11: Counterfactual Policy Designs (reprinted)



Residual Fiscal Cost as a Function of the Exemption Percentile

14 CMED-Regulated Pharmaceuticals vs. Non-Pharma Medical Supplies

The pharma decomposition splits Group 65 into class 6531 (medications subject to CMED price ceilings) and the remaining classes (non-pharma medical supplies). Split-sample DiD

Table OA.12: Pharmaceuticals (CMED-Regulated) vs. Non-Pharma Medical Supplies

	Pharma (class 6531) split sample	Non-pharma (other 65) split sample	Interaction full sample
Log prices			
$g65 \times Pre$	-0.1573***	-0.0973***	-0.0988***
	(0.0117)	(0.0086)	(0.0087)
$g65 \times Pre \times$ Pharma	–	–	-0.0742***
	–	–	(0.0190)
Log firms			
$g65 \times Pre$	0.1661***	0.0504***	0.0510***
	(0.0099)	(0.0057)	(0.0057)
$g65 \times Pre \times$ Pharma	–	–	0.1081***
	–	–	(0.0114)
Log bids			
$g65 \times Pre$	0.1355***	0.0056	0.0058
	(0.0129)	(0.0081)	(0.0081)
$g65 \times Pre \times$ Pharma	–	–	0.1193***
	–	–	(0.0153)
Distance (km)			
$g65 \times Pre$	7.8602*	2.1939	1.9921
	(4.3790)	(1.8406)	(1.8420)
$g65 \times Pre \times$ Pharma	–	–	7.1415
	–	–	(4.6229)
Obs. (Log prices)	55,794	593,920	649,714
Obs. (Log firms)	67,647	705,616	773,263
Obs. (Log bids)	67,647	705,616	773,263
Obs. (Distance)	55,794	593,920	649,714
Item FE + PBU FE	YES	YES	YES
Implied fiscal cost (pre-period, nominal)			
Value of Group-65 procurement (R$M)	293.1	395.9	689.0
Fiscal cost (R$M)	42.7	36.7	79.3
Share of total cost	54%	46%	100%

Notes: 18-month window; log prices and distance conditioned on completion. Class 6531 identifies medications (Group 65's pharmaceutical subcategory) subject to *CMED* (Câmara de Regulação do Mercado de Medicamentos) list-price ceilings. Remaining Group-65 classes (non-pharma medical supplies) are unregulated. The first two columns estimate the headline DiD separately within each subsample. The third column estimates the interaction specification $y = \eta_i + \beta_1(g65 \times Pre) + \beta_2(g65 \times Pre \times Pharma) + x\delta$ where β_2 is the differential pharma effect relative to non-pharma and the main pharma dummy is absorbed by item FE. SEs clustered at the item level. The implied fiscal cost applies the class-specific $\hat{\beta}$ on log prices to the pre-period Group-65 procurement value within that class. *** p<0.01, ** p<0.05, * p<0.1.

estimates are −0.1573 for pharma and −0.0973 for non-pharma; the differential is confirmed by an interaction specification within the full 18-month sample ($\hat{\beta}_{\text{g65\_pre} \times \text{pharma}} = -0.074$, $p < 0.01$). Pharma bears 54% of the total fiscal cost despite only 43% of the procurement value, because the per-unit distortion is 62% larger. Implementation details are in `scripts/24_pharma.R`.

15 Real Prices (IPCA-Deflated)

Table OA.13: Real Prices (IPCA-deflated, log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.1077***	-0.1210***	-0.0994***	-0.0997***	-0.0762***	-0.0795***
	(0.0121)	(0.0117)	(0.0108)	(0.0104)	(0.0096)	(0.0094)
Sealed bids	-0.1464***	-0.2225***	-0.1422***	-0.2029***	-0.1490***	-0.2059***
	(0.0088)	(0.0131)	(0.0079)	(0.0123)	(0.0079)	(0.0116)
lquantity	-0.2603***	-0.2978***	-0.2622***	-0.2966***	-0.2609***	-0.2944***
	(0.0082)	(0.0089)	(0.0075)	(0.0080)	(0.0073)	(0.0077)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1911	0.2128	0.1936	0.2115	0.1941	0.2112
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Constant absorbed by fixed effects.

16 Extensive Margin and Procurement Efficiency

17 Heterogeneity by PBU Type

18 SME Winner Composition

Figure OA.10 displays the monthly share of completed items where the winning firm is self-classified as an SME.

Table OA.14: Extensive Margin: Tender Completion Rate

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	0.1165***	0.1179***	0.1244***	0.1265***	0.1072***	0.1133***
	(0.0044)	(0.0044)	(0.0035)	(0.0036)	(0.0032)	(0.0032)
Sealed bids	0.0735***	0.0607***	0.0638***	0.0486***	0.0651***	0.0544***
	(0.0025)	(0.0033)	(0.0020)	(0.0025)	(0.0018)	(0.0023)
lquantity	0.0224***	0.0256***	0.0258***	0.0284***	0.0251***	0.0274***
	(0.0007)	(0.0007)	(0.0006)	(0.0006)	(0.0005)	(0.0005)
Observations	304,392	304,392	609,299	609,299	901,506	901,506
R-squared	0.0181	0.0177	0.0211	0.0209	0.0195	0.0194
Group FE	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: DV = 1 if item was successfully purchased, 0 otherwise. All items included (not filtered to completed). Group FE used instead of item FE. Standard errors clustered at the item level. *** p<0.01, ** p<0.05, * p<0.1.

Table OA.15: Price Efficiency: Final Price / Reference Price (%)

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.0651***	-0.0688***	-0.4399	-0.5009	-0.3033	-0.3462
	(0.0115)	(0.0109)	(0.3740)	(0.4354)	(0.2566)	(0.3010)
Sealed bids	-0.0620***	-0.0501***	0.0039	0.0605	-0.0218	0.0113
	(0.0027)	(0.0044)	(0.0608)	(0.1062)	(0.0371)	(0.0623)
lquantity	-0.0148***	-0.0155***	0.1277	0.1382	0.0755	0.0804
	(0.0018)	(0.0018)	(0.1426)	(0.1521)	(0.0909)	(0.0962)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.0038	0.0025	0.0001	0.0001	0.0001	0.0001
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Constant absorbed by fixed effects.

Table OA.16: Heterogeneous Effects by PBU Type (18-month window)

	Log prices		Log firms	
	Base	+PBU FE	Base	+PBU FE
$g65 \times Pre$	-0.0718***	-0.0697***	-0.0146	0.0567***
	(0.0231)	(0.0230)	(0.0113)	(0.0120)
Direct admin.	0.0425***	–	0.0421***	–
	(0.0084)		(0.0044)	
$g65 \times Pre\times$ Direct admin.	-0.0655***	-0.0685***	0.1175***	0.0475***
	(0.0236)	(0.0227)	(0.0117)	(0.0122)
Observations	649,714	649,714	773,263	773,263
R-squared	0.1936	0.2108	0.2065	0.1632
Item FE	YES	YES	YES	YES
PBU FE	NO	YES	NO	YES

Notes: 18-month window. Direct admin. is 1 for PBUs under direct administration, 0 otherwise. Standard errors clustered at the item level. *** p<0.01, ** p<0.05, * p<0.1.

Table OA.17: Winner Composition: SME Winner (binary)

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.3388***	-0.3615***	-0.3633***	-0.3771***	-0.3858***	-0.4056***
	(0.0053)	(0.0053)	(0.0044)	(0.0044)	(0.0040)	(0.0039)
Sealed bids	0.4126***	0.3882***	0.4614***	0.4341***	0.4190***	0.3918***
	(0.0043)	(0.0053)	(0.0037)	(0.0039)	(0.0032)	(0.0032)
lquantity	0.0096***	0.0141***	0.0090***	0.0126***	0.0062***	0.0104***
	(0.0010)	(0.0008)	(0.0008)	(0.0006)	(0.0007)	(0.0006)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1410	0.1267	0.1746	0.1486	0.1583	0.1333
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Constant absorbed by fixed effects.

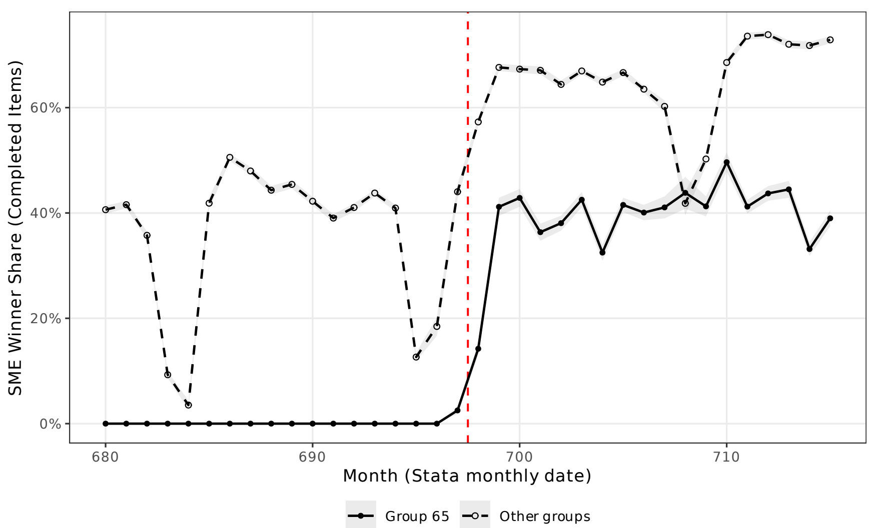


Figure OA.10: SME Winner Share (Completed Items) by Month

Part IV

RAIS Matched Employer-Employee Linkage

Brazil's *Relação Anual de Informações Sociais* (RAIS) is a matched employer-employee registry covering the universe of formal-sector employment relationships. This part links the winning firms in BEC to their RAIS profiles at the CNPJ-raiz (8-digit firm identifier) level, using the establishment-level (ESTB) files for 2011–2017. Of 17,857 distinct winning CNPJ raízes in the Paper-2 sample, 14,755 match RAIS 2017 nationwide (82.6% coverage); within Sao Paulo state alone, 12,533 match (70.2%). The link is produced by `scripts/05_link_rais.py`, a DuckDB-based ETL.

19 Validation of the Self-Declared ME/EPP Label

The BEC data includes a self-declared `fornec_enquad` (firm-enquadramento) variable. Table OA.18 tests whether the headline DiD is sensitive to the label's classification noise by restricting the estimation sample to firms with (i) a RAIS match, (ii) active employment at most 49 (ME/EPP ceiling), (iii) active employment at most 9 (micro ceiling), and (iv) active employment strictly below 100 (excluding the top of the RAIS size distribution). The price coefficient is essentially invariant across these subsamples.

Table OA.18: RAIS-Validated Winner Subsamples

20 Winner Composition: RAIS-Validated SME

Table OA.19 uses a RAIS-validated SME flag (winner active employment $\leq$ 49) as the dependent variable. The DiD on this outcome estimates how the SME-only rule shifts the probability of a genuinely small winner; the mechanism prediction is a positive post-policy shift.

Table OA.19: Winner Composition: RAIS-Validated SME

Table OA.20: Winner Composition: Alternative RAIS Size Thresholds (18-month window)

	Micro ($\leq$9 vínculos)		Not large ($<$100 vínculos)	
	Base	+PBU FE	Base	+PBU FE
$g65 \times Pre$	-0.2511***	-0.2637***	-0.1298***	-0.1362***
	(0.0040)	(0.0041)	(0.0041)	(0.0042)
Observations	649,714	649,714	649,714	649,714
R-squared	0.0212	0.0217	0.0097	0.0096
Item FE	YES	YES	YES	YES
PBU FE	NO	YES	NO	YES

Notes: Item-level binary outcome equal to one if the winner's CNPJ raiz matches RAIS 2017 (Brasil) with at most 9 (Micro) or fewer than 100 (Not large) active employment links summed across establishments. 18-month window, completed items. Compare to Table **??** (SME $\leq$49) and Table OA.17 (self-declared). Standard errors clustered at the item level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

21 Main DiD with RAIS Firm-Level Controls

Table OA.21 augments the headline DiD with RAIS firm-level controls—log(1 + vínculos), log(1 + firm age at 2018), and CNAE 2-digit fixed effects. The $g65 \times Pre$ coefficient is invariant, confirming that the price effect is not explained by observable differences in winner size, age, or sector.

Table OA.21: Main DiD with RAIS Firm-Level Controls

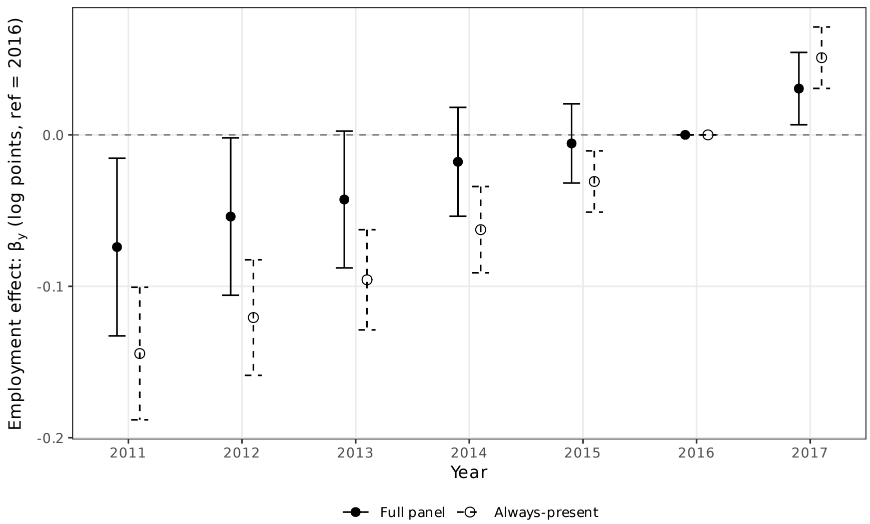
22 Enforcement Heterogeneity

Table OA.22 tests whether the price effect is larger in PBUs that were historically more exposed to non-SME winners. The exposure measure is the pre-period share of each PBU's winners with RAIS employment ≥ 100. Weighted median is 6.9%; PBUs above median show a -0.049 ($p < 0.05$) larger price distortion.

Table OA.22: Enforcement Heterogeneity by PBU Exposure

23 Firm-Level Employment Pre-Trends (RAIS 2011–2017)

Figure 23 plots the event-study coefficients on log(1 + vínculos) for treated firms (ever won a group-65 item in the BEC pre-period) versus controls (won only non-group-65 items in the same period). Pre-reference joint F-test is 1.46 ($p = 0.198$) in the balanced panel with zero-fill, failing to reject parallel pre-trends. The always-present subset (firms with ≥ 1 employment link every year 2011–2017) rejects more strongly; this reflects the selection of procurement winners on firm productivity and growth rather than a violation of the paper's item-level identification.



Firm-Level Employment Pre-Trends (RAIS 2011–2017)

Table OA.23: Pre-Trends Event Study

24 Firm Age Heterogeneity

Table OA.24 interacts $g65 \times Pre$ with a young-firm indicator (age at 2018 $\leq$ 9 years, the median of pre-period winners). The price effect is uniform across age within the ME/EPP-eligible pool; the apparent differences reported by naive age splits are driven by firm-size composition (see Section OA.27).

Table OA.24: Heterogeneity by Winner Firm Age

25 CNAE 2-Digit Sector Heterogeneity

Table OA.25 re-estimates the DiD within each of the top-6 CNAE 2.0 divisions (wholesale 46, retail 47, vehicle wholesale 45, chemicals 20, food manufacturing 10, other manufacturing 32) and reports the coefficient on $g65 \times Pre$.

Table OA.25: Heterogeneity by Winner CNAE 2-Digit Sector

26 Winner Age × Size Quadrants

Table OA.26 crosses winner age (young/mature, median split at 9 years) with RAIS-observed size (SME/large, cut at the 49-vínculos ME/EPP ceiling) to form four quadrants. Within eligible firms (SME), the price effect is essentially identical for young and mature firms (-0.129 vs -0.128)—the rent transfer from the SME-only rule is uniform within eligibility and concentrated on the geographic/competition margins rather than on firm maturity per se.

Table OA.26: Heterogeneity by Age × Size Quadrants

Part V

Complementary Estimators

27 Synthetic Control

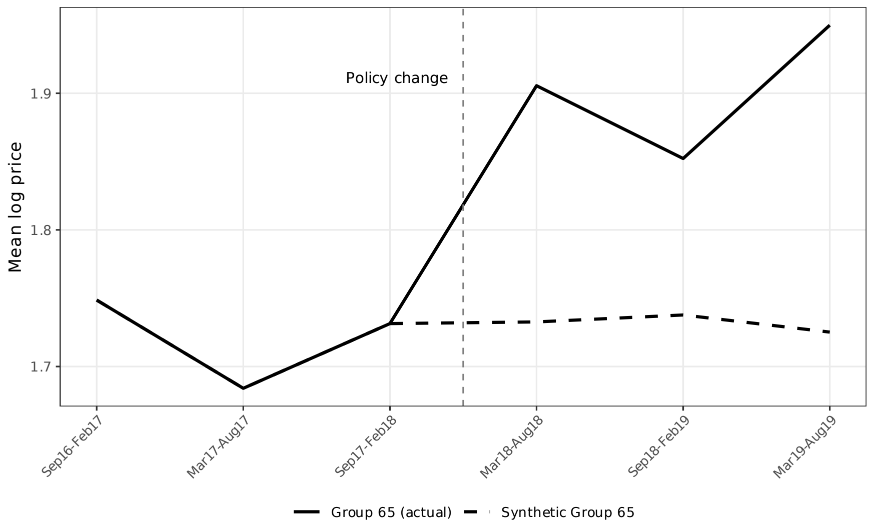


Figure OA.11: Synthetic Control: Group 65 (Actual) vs. Synthetic Group 65

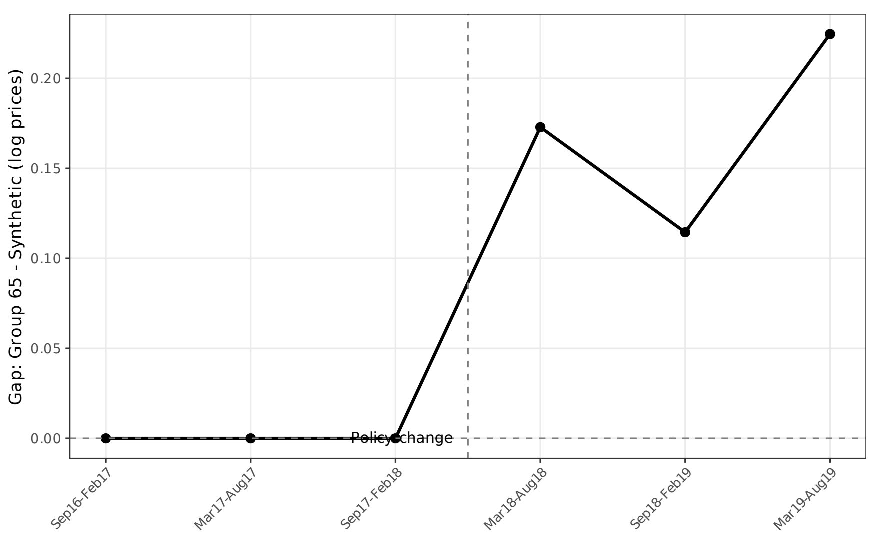


Figure OA.12: Synthetic Control: Gap Between Group 65 and Synthetic (Log Prices)

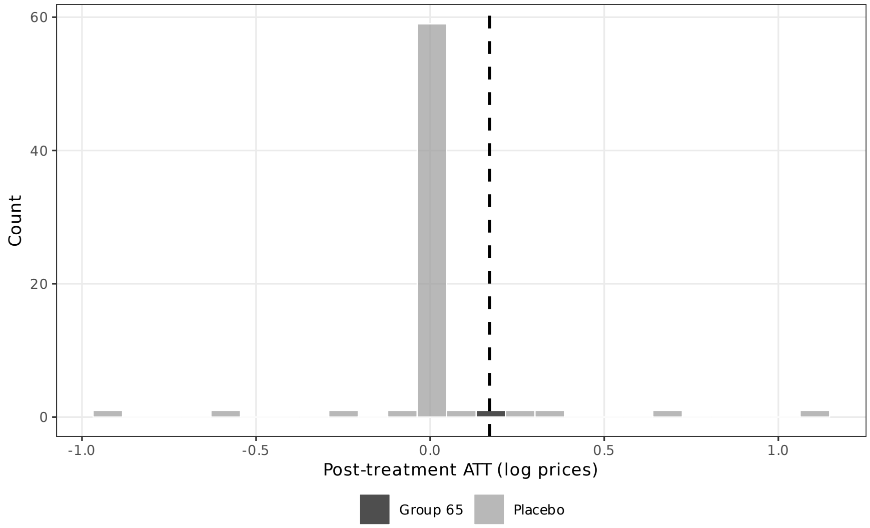


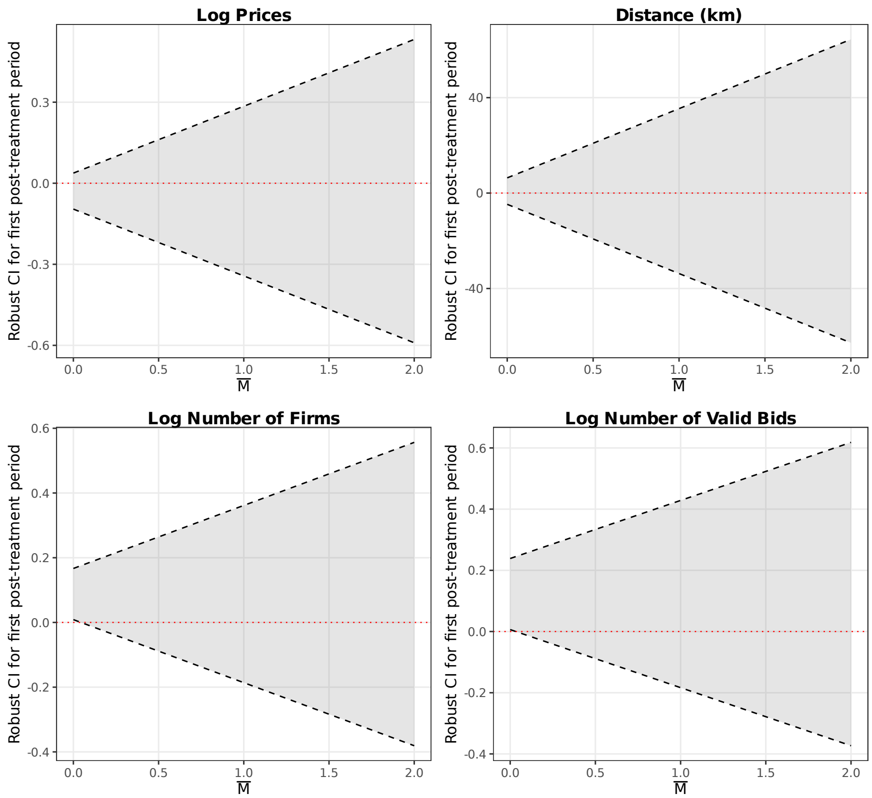
Figure OA.13: Synthetic Control: In-Space Placebo Distribution of Post-Treatment ATTs

Table OA.27: Synthetic Control: Summary Results

Panel A: Gaps	
Mean pre-treatment gap (semesters 1–3)	0.0000
Mean post-treatment gap (semesters 4–6)	0.1707
Post-treatment ATT (mean)	0.1707
Placebo rank p-value	0.103 (7/68)
L2 imbalance	0.0000
Panel B: Top donor weights (> 1%)	
Product group	Weight
Group 40	0.044
Group 75	0.099
Group 79	0.382
Group 89	0.436
Other groups	0.039
Balanced groups	69
Semesters	6

Notes: Augmented synthetic control with Ridge augmentation. Outcome: mean log price at the group × semester level. Treatment date: March 2018 (group 65 loses open-tender exemption). Pre-treatment gap measures how much lower group 65 prices were under open tenders relative to the synthetic counterfactual. Post-treatment gap near zero validates the parallel trends assumption. Inference via conformal method.

28 HonestDiD Sensitivity Analysis



HonestDiD Sensitivity Analysis of the Log-Price Effect

29 Lee (2009) Bounds

Table OA.28: Lee (2009) Bounds: Sample Selection Correction

	Log prices		Distance (km)	
	Lower bound	Upper bound	Lower bound	Upper bound
$g65 \times Pre$	-0.1306***	-0.1227***	10.7750***	10.8042***
	(0.0096)	(0.0085)	(2.2322)	(2.2328)
Observations	625,816	625,814	625,814	625,814
R-squared	0.1870	0.1941	0.0009	0.0013
Trimming proportion		0.0882		
Item FE	YES	YES	YES	YES

Notes: Lee (2009) bounds correct for sample selection arising from treatment effects on completion rates. 18-month window. Lower/upper bounds obtained by trimming the outcome distribution in the excess-selected cell. DiD in completion rate: -0.0882. Standard errors clustered at the item level. *** p<0.01, ** p<0.05, * p<0.1.

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Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics*, 225(2): 200–230.

Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics*, 225(2): 254–277.

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